Thermodynamic Feasibility of a Solar-Powered EV Charging Station in Three Indian Climates: An Hourly First- and Second-Law Analysis

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***Abstract*—One fixed solar-plus-storage EV charging station (49.6 kWp TOPCon PV, 93.2 kWh LFP battery, 222 kWh/day load) is simulated hour by hour over three years of satellite weather (NASA POWER 2021–2023) in Jodhpur, Bengaluru and Guwahati. The design is identical in every city, so climate, tariff and grid emission factor are the only differences. The model closes the First Law and the exergy balance every hour to 10^{-13} kWh. Cell temperature comes from a convection–radiation balance calibrated to the datasheet NOCT, efficiency includes a low-light term, and the solar exergy input is computed per irradiance component: Petela's factor for the beam, and diluted-radiation theory for diffuse and ground-reflected light. The EV load timing is taken from 24,285 measured public-charging sessions. Solar covers 79.7 %, 74.3 % and 70.4 % of the EV demand in Jodhpur, Bengaluru and Guwahati. The PV cell accounts for about 98 % of all exergy destroyed, about 80 % of it inside the cell. The Second Law gives a result an energy balance cannot: in cloudy, diffuse-light conditions the PV energy efficiency falls but its exergy efficiency rises, because diffuse light carries 0.67–0.80 of its energy as exergy against 0.93 for the beam. As a result Jodhpur, the sunniest city, has the lowest exergy efficiency (24.8 % against 25.9 % in Bengaluru). These rankings hold in all three weather years and in all 200 paired Monte Carlo draws. The coverage ranking reverses on ERA5 reanalysis data, so the choice of weather source matters more than any model parameter. PV alone generates below the EV tariff in every city; adding the battery makes the net present value negative everywhere at current prices.**

***Index Terms*—Exergy analysis; photovoltaics; EV charging; LFP battery; diluted radiation; Petela; Monte Carlo; India**

I. Introduction

India's electric-vehicle fleet is growing faster than its public charging network. Where chargers are built, most draw from a grid whose weighted-average emission factor is 0.710 tCO_2/MWh [28]. Pairing chargers with on-site PV and a battery is the obvious response. But a charging station is a load that runs into the evening, while PV produces at midday. How much of the load the sun can actually meet therefore depends on timing and storage as much as on how sunny the site is.

This project asks three questions of one fixed station design:

1) How much of a fixed EV charging demand can solar cover, hour by hour, in three climatically different Indian cities?
2) Where, by the Second Law, is the work potential of the incoming sunlight destroyed, and how does that change with climate and season?
3) Does the economic verdict follow the thermodynamic one?

The three cities span the axes that affect PV performance. **Jodhpur** (26.3°N) is hot, arid and beam-rich. **Bengaluru** (13.0°N) is mild, with two monsoons and the highest EV density of the three. **Guwahati** (26.2°N) is humid and has a heavy south-west monsoon.

It is a feasibility study, not an optimisation. The dispatch rule is a fixed priority order, the station size is fixed, and economics and CO_2 are a reporting layer computed from the simulated flows.

II. Related Work

- **PV exergy methodology.** Reviews of exergy analysis for solar electricity [17] report PV exergy efficiencies of roughly 10–20 %. Most use Petela's factor [1] for *all* incident light. Diffuse radiation is diluted: it has more entropy per joule and therefore less exergy. Landsberg and Tonge [2] give the thermodynamics of diluted radiation; this work applies it per irradiance component.
- **Cell temperature.** Steady and transient energy-balance models with convection and long-wave radiation [11], [12] predict module temperature better than the NOCT correlation [5], [13], especially with wind data. Wind correlations of the Watmuff type [10] and the Swinbank sky temperature [9] are standard ingredients.
- **PV efficiency.** The PVGIS power-rating model of Huld et al. [3] adds irradiance and temperature dependence to crystalline-silicon output. It was recently refitted for modern modules [4].
- **Solar EV charging in India.** A 10 kW bifacial PV + $LiFePO_4$ station built in Gorakhpur [18] uses the same chemistry and a priority-based EMS. Multi-city techno-economic studies [19], [20] find that cost and payback shift with climate and that Jodhpur ranks among the most cost-effective sites.

Gap. No study found combines (i) a full hourly exergy balance for a solar EV station, (ii) component-wise solar exergy, and (iii) a multi-city Indian climate comparison with the design held fixed. This project combines them.

III. Station Design and Input Data

A. Fixed design

Table I lists the design. Everything except the weather, the tariff and the grid emission factor is the same in all three cities.

Sizing logic. Annual EV energy is $222 \times 365 = 81{,}030$ kWh. India's typical specific yield is ≈1,630 kWh/kWp, which gives ≈49.7 kWp, or 80 modules of 0.620 kWp. The battery is two standard REPT Y52 modules. At 0.5C it accepts 46.6 kW, close to the array peak, so midday surplus is rarely clipped by battery power. These are reference sizes, not optima; Section VI-D sweeps both.

Orientation. Due south maximises annual irradiation in the northern hemisphere. One tilt is used everywhere so that geometry is not a second variable. Re-transposing the NASA POWER data at 0–40° shows the annual optimum at 28° (Jodhpur, Guwahati) and 14° (Bengaluru). The 20° compromise loses only 0.67 %, 0.86 % and 0.40 % of plane-of-array irradiation. Section V gives the check.

Series string. Wiring the 104 cells as one series string (1P104S) does not change the cells' own I^2R loss. At a fixed power, a 2P52S layout halves the voltage and the pack resistance falls to a quarter, but the current doubles, so the cell loss is identical. Series is chosen because 333 V sits inside common PCS voltage windows, external cable current is halved, and no parallel strings need matching. The current at 46.6 kW is 140 A, the cell's rated 0.5C standard current.

B. Weather data: three years, two sources

Main case: NASA POWER 2021–2023 [21]. This is hourly satellite irradiance (CERES SYN1deg: GHI, DNI, DHI) with MERRA-2 2 m temperature and 10 m wind. The horizontal values are transposed to the 20° plane with the Hay–Davies model [5], [6] and a ground albedo of 0.2 (Appendix A-F). Each POWER value is a mean over the UTC hour; the centre of the daily irradiance curve sits 0.42–0.51 h before solar noon, which confirms this. The rows are therefore stamped at hh:30 UTC and shifted to IST. The three years run back to back as one 26,280-hour simulation, so the battery state carries over between years. All annual numbers are 3-year means unless a year is named.

Cross-check: PVGIS-ERA5 2023 [22]. These files come already transposed to 20°/180°. ERA5 is a reanalysis, not a satellite product. It gives Jodhpur only ≈400 W/m^2 at noon on clear January days, against 900–1,000 W/m^2 from clear-sky physics and the satellite data.

C. EV load: measured timing, fixed energy

The Checkpoint-2 profile (2/15/5 kW for night/day/evening) was an assumption that put 81 % of the energy in daylight. It is replaced by the hourly shape of real sessions from an open dataset of 72,856 transactions at 2,119 Korean chargers in 2022 [23]. No comparable session-level dataset is open for India. The main case uses public-area, public-institution and public-parking chargers (24,285 sessions). Each session's energy is spread evenly over its duration and binned by clock hour, then the 24 shares are normalised and scaled to 222 kWh/day. Only the *timing* comes from the data; the daily energy is unchanged, so the comparison with Checkpoint 2 stays like-for-like.

The measured public profile puts 56 % of the energy between 06:00 and 18:00, against 81 % in the synthetic one. It peaks at 12.8 kW at 19:00 (Fig. 2). Workplace, residential and synthetic profiles are kept as sensitivity cases.

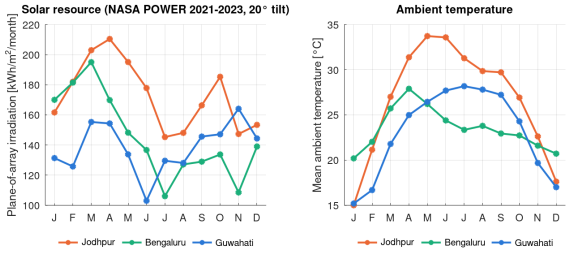


Fig. 1. Monthly plane-of-array irradiation and ambient temperature, NASA POWER 2021–2023 mean.

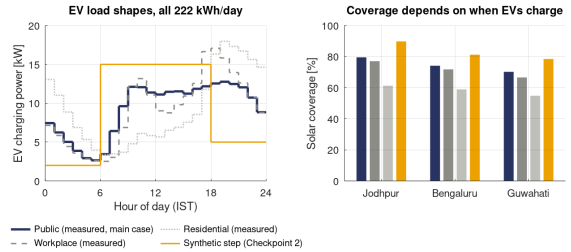


Fig. 2. EV load shapes (all 222 kWh/day) and their effect on solar coverage.

IV. Methodology

A. System boundary and pipeline

The station is treated as three open control volumes inside one boundary:

1) **CV1:** PV laminate.
2) **CV2:** battery pack.
3) **CV3:** power electronics and wiring.

Sun, grid, EV and ambient air are boundary conditions. MPPT is a black box that delivers maximum power.

The MATLAB/Octave pipeline is `params`→`load_weather`→`simulate_station` [`pv_model`→`ems_dispatch`→`battery_model`, hourly]→`postprocess`→`economics`.

Every annual figure is a sum or a ratio of sums of hourly outputs; weather is never averaged first. Because $\eta(G, T_c)$ and $Ex(G, T_0)$ are nonlinear, averaging the inputs first would bias the results (Jensen's inequality).

TABLE I
FIXED STATION DESIGN (IDENTICAL IN ALL CITIES)

Item	Value	Source / basis
PV module	Trina Vertex N TSM-NE19R 620 W (N-type TOPCon), 2.382×1.134 m, $\eta_{ref} = 620/(2.701 \times 1000) = 22.95\,\%$, $\beta = -0.29\,\%$/K, NOCT $43 \pm 2\,^{\circ}$C	Datasheet DT-M-0093-B (2024) [24]
PV array	80 modules $= 49.6$ kWp, $A = 216.1\,\text{m}^2$, tilt 20°, azimuth 180° (due south)	Sized so annual yield $\approx$ annual demand: $81{,}030/1{,}630 \approx 49.7$ kWp
Battery	REPT 280 Ah LFP cells, 104 in series (2 Y52 modules, 1P104S), $104 \times 3.2 \times 280 = 93.2$ kWh, 332.8 V, $R = 104 \times 0.30\,\text{m}\Omega = 31.2\,\text{m}\Omega$	REPT ESS brochure [25]
Battery limits	SOC 10–90 % (74.6 kWh usable), 0.5C $= 46.6$ kW, 8,000 cycles to 70 % SOH	Brochure; SOC window is an engineering assumption
Converters	PV inverter $\nu_{inv} = 0.97$; battery PCS $\nu_{pcs} = 0.97$ each direction	Fixed coefficients from the 96–98 % range in the brief
DC wiring	2 % lumped ohmic + mismatch loss	Industry defaults [26], [27]
EV load	222 kWh/day (81.03 MWh/yr) at the AC bus; hourly shape from measured public sessions; the same every day and in every city	Daily energy is a design value; timing is measured [23]
EMS	PV→EV; surplus→battery→export; deficit←battery←grid; battery charged from PV only	Priority rule, not optimised

TABLE II
ANNUAL PLANE-OF-ARRAY IRRADIATION [KWH/M^2]

	Jodhpur	Bengaluru	Guwahati
NASA POWER 2021	2,086	1,703	1,669
NASA POWER 2022	2,110	1,704	1,605
NASA POWER 2023	2,030	1,826	1,714
NASA POWER mean	**2,076**	**1,745**	**1,662**
PVGIS-ERA5 2023	1,598	1,774	1,300
Diffuse + ground share (POWER)	37 %	52 %	48 %
Mean ambient T_a [°C]	26.7	23.5	23.1

B. Solar geometry and optical losses

The angle of incidence θ on the tilted plane comes from standard relations [5]. They are evaluated at the UTC mid-hour of each row and use the equation of time:

$$\delta = 23.45^{\circ} \sin\left[\tfrac{360}{365}(284 + n)\right], \tag{1}$$

$$\omega = 15^{\circ}(t_{solar} - 12), \tag{2}$$

$$\cos\theta = \sin\delta \sin(\phi - \beta) + \cos\delta \cos(\phi - \beta) \cos\omega. \tag{3}$$

Here n is the day of year, ϕ the latitude, $\beta = 20^{\circ}$ the tilt, and t_{solar} the apparent solar time in hours.

Reflection at the glass is applied to each component separately:

$$G_{\text{eff}} = G_b\,\text{IAM}_b(\theta) + 0.95\,G_d + 0.65\,G_r, \tag{4}$$

$$\text{IAM}_b = 1 - b_0\left(\frac{1}{\cos\theta} - 1\right). \tag{5}$$

- The beam uses the ASHRAE modifier with $b_0 = 0.05$ [8], the PVsyst default.
- 0.95 is an effective sky-diffuse modifier at 20° tilt [7].
- 0.65 for ground light is an engineering estimate. Ground light is under 1 % of the plane irradiance at 20°, so its effect is negligible.

C. First Law: PV cell temperature (CV1)

Start from the rate form of the First Law for the laminate: absorbed radiation in, electrical work out, heat rejected by convection from both faces and long-wave radiation to sky and ground:

$$\begin{aligned} mc_p\frac{dT_c}{dt} = \alpha A G_{\text{eff}} - P - 2h(v)A(T_c - T_a) \\ - \varepsilon_f \sigma A(T_c^4 - T_{sky}^4) \\ - \varepsilon_b \sigma A(T_c^4 - T_a^4). \end{aligned} \tag{6}$$

A module's thermal time constant is 5–10 min, much shorter than the 1 h step, so $dT_c/dt \approx 0$. Equation (6) is then solved for T_c each hour by Newton iteration, coupled to $P = \eta(T_c)AG_{\text{eff}}$: a cooler cell is more efficient, and every watt exported as electricity is a watt that does not heat the cell.

The parameters are:

- $\alpha = 0.90$, the fraction of transmitted light absorbed [5].
- $\varepsilon_f = 0.84$ (glass) and $\varepsilon_b = 0.89$ (backsheet), within published ranges of 0.84–0.91 [14].
- $T_{sky} = 0.0552\,T_a^{1.5}$ (Swinbank [9]).
- $h(v) = h_{free} + 3.0\,v$ per face, with the wind slope from Watmuff et al. [10].
- $v = v_{10}(3/10)^{1/7}$: the 10 m wind scaled to the 3 m canopy height with the 1/7 power law.

Calibration of h_{free}. h_{free} is not assumed. It is the value that makes (6) return the datasheet NOCT exactly at the NOCT

test conditions (800 W/m^2, $T_a = 20\,^\circ$C, 1 m/s, open circuit so $P = 0$):

$$h = \frac{\alpha G - \varepsilon_f \sigma(T_N^4 - T_{sky}^4) - \varepsilon_b \sigma(T_N^4 - T_a^4)}{2(T_N - T_a)}, \tag{7}$$

$$h_{free} = h - 3.0 \times 1 = 5.55 \text{ W/m}^2\text{K}. \tag{8}$$

With $T_N = 316.15$ K, $T_a = 293.15$ K and $T_{sky} = 277.1$ K: absorbed 720 W/m^2, radiation 195.2 + 131.5 = 326.7 W/m^2, so $h = 393.3/46 = 8.55$ per face and $h_{free} = 5.55$. In the Monte Carlo it is recalibrated for every NOCT draw.

The plain NOCT correlation $T_c = T_a + (\text{NOCT} - 20)G/800$ is this balance linearised at that one test point (Appendix A-D). It ignores wind, sky radiation and the cooling by electrical output, and it runs cells 6–9 K hot on an annual basis. It is kept as a sensitivity case.

D. First Law: PV efficiency and power

$$\eta = \eta_{ref}\, f_{low}(G')\,[1 - \beta(T_c - 25)], \quad G' = \frac{G_{\text{eff}}}{1000}, \tag{9}$$

$$f_{low} = 1 + k_1 \ln G' + k_2 (\ln G')^2, \tag{10}$$

$$k_1 = -0.017237, \qquad k_2 = -0.040465, \tag{11}$$

$$P_{DC} = \eta A G_{\text{eff}}, \qquad P_{AC} = P_{DC}(1 - 0.02)\,\nu_{inv}. \tag{12}$$

- **The temperature term** is the standard linear derating [13]. 25 °C is the STC cell temperature at which η_{ref} and β are measured, so the bracket equals 1 at the datasheet point, falls below 1 for hotter cells and rises above 1 for colder ones.
- f_{low} is the irradiance part of the Huld model in its PVGIS-5 crystalline-silicon fit [3]. It equals 1 at 1,000 W/m^2, 0.98 at 400 W/m^2 and 0.92 at 200 W/m^2.
- **Hybrid choice.** Huld's temperature terms (k_3–k_6) are dropped and the datasheet β is kept, because Huld's coefficients were fitted to older modules (about -0.47 %/K). The model is therefore "Huld irradiance terms + datasheet temperature coefficient", not the full Huld model.
- **Power chain.** P_{DC} is the definition of efficiency rearranged. The losses are multiplied, not added, because each acts on what remains after the previous one ($0.98 \times 0.97 = 0.9506$).

E. Second Law: exergy of the incident radiation

For blackbody radiation at T_s against an environment at T_0, the energy flux is σT^4 and the entropy flux is $\frac{4}{3}\sigma T^3$. Taking exergy relative to the environment's own radiation gives Petela's factor [1] (Appendix A-A):

$$\psi_P = \frac{Ex}{E} = 1 - \frac{4}{3}\frac{T_0}{T_s} + \frac{1}{3}\left(\frac{T_0}{T_s}\right)^4 \approx 0.931 \tag{13}$$

for $T_0 = 300$ K and $T_s = 5778$ K.

ψ_P describes undiluted light from the sun's disc, which is the direct beam. Diffuse sky light and ground-reflected light have roughly the solar spectrum spread over a far larger solid angle. This is *diluted* radiation, with spectral radiance $\varepsilon B_\nu(T_s)$, $\varepsilon \ll 1$. Its entropy flux per joule is higher [2]:

$$\psi(\varepsilon) = 1 - \frac{4}{3}\frac{T_0}{T_s} X(\varepsilon), \tag{14}$$

$$X(\varepsilon) = \frac{45}{4\pi^4 \varepsilon} \int_0^\infty x^2 \big[(1+n)\ln(1+n) - n \ln n\big] dx, \tag{15}$$

$$n = \frac{\varepsilon}{e^x - 1}. \tag{16}$$

$X(1) = 1$, so undiluted light recovers ψ_P (apart from the 10^{-5} environment term). The dilution is the radiance the module sees relative to the sun's surface:

$$\varepsilon_d = \frac{2G_d}{\sigma T_s^4 (1 + \cos\beta)}, \qquad \varepsilon_r = \frac{2G_r}{\sigma T_s^4 (1 - \cos\beta)}. \tag{17}$$

Typical ε_d is 10^{-6}–10^{-5}, giving $\psi \approx 0.67$–0.71; even bright overcast light stays below 0.80. X is integrated numerically once and tabulated. The component-wise solar exergy input is

$$Ex_{in} = A\,[G_b \psi_P + G_d \psi(\varepsilon_d) + G_r \psi(\varepsilon_r)], \tag{18}$$

with the dead state T_0 equal to the hourly ambient temperature.

F. Second Law: PV exergy balance and efficiencies

Electricity is pure exergy, so $Ex_{out} = P$. The PV loss $Ex_{in} - P$ is split into three parts:

$$Ex_{opt} = A \sum_k G_k (1 - \alpha\,\text{IAM}_k)\psi_k, \tag{19}$$

$$Ex_Q = Q_{heat}\left(1 - \frac{T_0}{T_c}\right), \tag{20}$$

$$Ex_{int} = Ex_{in} - P - Ex_{opt} - Ex_Q = T_0\, S_{gen,int} \geq 0. \tag{21}$$

$Ex_{int} \geq 0$ is asserted every hour.

- Ex_{opt}: reflected or unabsorbed light.
- Ex_Q: the work potential of the heat rejected at T_c, which is lost to the surroundings.
- Ex_{int}: destruction inside the cell (thermalisation of hot carriers, recombination, resistive loss), via Gouy–Stodola [15].

The efficiencies are

$$\eta_{PV} = \frac{P}{AG}, \quad \psi_{PV} = \frac{P}{Ex_{in}} \;\Rightarrow\; \frac{\eta_{PV}}{\psi_{PV}} = \frac{Ex_{in}}{AG}. \tag{22}$$

This identity drives the main Second-Law result: ψ_{PV}/η_{PV} is set by the exergy content of the light, so a more diffuse sky raises ψ_{PV} even when η_{PV} falls. Because $Ex_{in} < AG$, $\psi_{PV} > \eta_{PV}$ always. (The Checkpoint-2 slides stated $\psi < \eta$; this was wrong and is corrected here.)

G. Battery (CV2): lumped electrical, thermal and Second Law

$$I = \frac{P_{term}}{V_{nom}}, \qquad \dot{Q} = I^2 R, \tag{23}$$

$$E_{t+1} = E_t + (P_c - P_d - I^2 R)\Delta t, \tag{24}$$

$$m c_p \frac{dT_b}{dt} = I^2 R - UA(T_b - T_a). \tag{25}$$

The thermal equation is solved exactly over each step. The Joule heat is generated at T_b and finally rejected at T_0. Internal generation $\dot{Q}/T_b$ plus heat-transfer generation $\dot{Q}(1/T_0 - 1/T_b)$ sums to $\dot{S}_{gen} = \dot{Q}/T_0$, so

$$\dot{E}x_{dest,batt} = T_0 \dot{S}_{gen} = I^2 R. \tag{26}$$

Charge and discharge limits solve $P \mp kP^2 =$ (headroom or available energy)$/\Delta t$, with $k = R \cdot 1000/V^2$. The SOC window and the 0.5C cap therefore hold *including* the loss term (Appendix A-E).

Ageing is a fade of $(1 - 0.70)/8000 = 3.75 \times 10^{-5}$ per equivalent full cycle at 25 °C, scaled by an Arrhenius factor in T_b ($E_a = 31.7$ kJ/mol [16]). The thermal mass (104×5.5 kg, $c_p = 1100$ J/kgK) and the enclosure conductance $UA = 40$ W/K are assumptions.

H. Converters and wiring (CV3)

$$Ex_{dest,inv} = (1 - \nu_{inv}) P_{DC}, \qquad Ex_{dest,wire} = 0.02\, P_{raw}, \tag{27}$$

$$Ex_{dest,pcs} = (1 - \nu_{pcs}) P_{in} \text{ per pass.} \tag{28}$$

Electrical input and output are both pure exergy, so for these components $\psi = \eta$ and the energy loss equals the exergy destroyed. The battery path, PV→PCS→cells→PCS→EV, has an AC-to-AC round-trip efficiency of $\approx 0.97^2 \times 0.99 \approx 0.93$ in the simulation (the cell I^2R is about 0.8 % of throughput).

I. EMS dispatch and the station-level balances

Each hour the priority rule (Table I) sets the flows. The battery's terminal powers are DC, and the PCS losses sit between the bus and the battery. Two balances are asserted every hour:

$$AG + P_{imp} = P_{EV} + P_{exp} + \Delta E_b + E_{opt} + Q_{heat} + L_{wire} + L_{inv} + L_{pcs} + I^2 R, \tag{29}$$

$$Ex_{in} + P_{imp} = P_{EV} + P_{exp} + \Delta E_b + \textstyle\sum_i Ex_{dest,i}. \tag{30}$$

Coverage, also called the solar contribution ratio (SCR), and the station efficiencies are

$$\mathrm{SCR} = \frac{E_{PV \to EV} + E_{batt \to EV}}{E_{EV}}, \qquad \mathrm{GD} = \frac{E_{imp}}{E_{EV}}, \tag{31}$$

$$\psi_{solar} = \frac{E_{EV} - E_{imp} + E_{exp} + \Delta E_b}{Ex_{in}}, \tag{32}$$

$$\psi_{sys} = \frac{E_{EV} + E_{exp} + \Delta E_b}{Ex_{in} + E_{imp}}, \tag{33}$$

where GD is the grid dependency.

J. Economics and CO_2 (reporting layer)

All values are in real INR with an 8 % real discount rate d over a 25-year life L. Module degradation is 1 % in year 1, then 0.4 %/yr, from the datasheet. The cost per solar kWh used is

$$\mathrm{LCOE} = \frac{C_0 + \sum_y (OM_y + R_y)(1+d)^{-y}}{\sum_y E_{solar} D_y (1+d)^{-y}}, \tag{34}$$

TABLE III
VALIDATION CHECKS

Check	Result	Status
Hourly First-Law closure (26,280 h×3)	max 5.7×10^{-14} kWh	Pass (asserted)
Hourly exergy closure	max 5.7×10^{-14} kWh	Pass (asserted)
$Ex_{int} \geq 0$ every hour	min 0 (night)	Pass (asserted)
Sun elevation vs PVGIS H_{sun}	mean error $-0.2°$, max 1.2°	Pass
POWER hour stamps vs solar noon	offset 0.42–0.51 h	Pass (hour means)
T_c model at NOCT conditions	43.0 °C	Pass (by construction)
Peak T_c, Jodhpur	67.1 °C	In 55–70 °C field range
Specific yield (DC)	1,508–1,883 kWh/kWp	Indian range
Performance ratio (AC)	0.862–0.867	High (no soiling modelled)
$X(\varepsilon)$ table vs direct integration	4 decimals at $\varepsilon = 1 \ldots 10^{-7}$	Pass
Tilt re-transposition at 20°	reproduces 2,076/1,745/1,662	Pass
Loss vs optimum tilt (28/14/28°)	0.67/0.40/0.86 %	Small
Sensitivity sweeps at baseline point	identical to baseline	Pass (asserted)
PVGIS PVcalc yield comparison	not run	Open

where E_{solar} is the solar energy served to EVs plus exports, and D_y is the degradation factor in year y. The NPV compares the station against a grid-only baseline:

$$\mathrm{NPV} = -C_0 + \sum_{y=1}^{L} \frac{D_y (E_{served}\tau + E_{exp} c_{exp}) - OM_y - R_y}{(1+d)^y}. \tag{35}$$

Here τ is the state energy tariff and $c_{exp} = 3$ INR/kWh the export credit.

Capex is PV 40,000 INR/kWp, battery 18,000 INR/kWh, inverter 6,000 INR/kW, PCS 8,000 INR/kW, plus 15 % balance of system, giving 49.8 lakh in total. O&M is 1.5 %/yr. The battery is replaced at year 15 at 60 % of today's price, and the converters at year 12. CO_2 uses the CEA national factor of 0.710 t/MWh [28], since India runs one integrated grid and state factors are not published.

V. VALIDATION

Table III lists the checks.

Literature comparison. The ψ_{PV} values of 24.8–25.9 % sit above the 10–20 % usually reported [17], for two reasons. The module is a 23 % TOPCon, against the 12–16 % modules in most of that literature. And most published values divide by Petela's factor for *all* light, which inflates Ex_{in} by 11–15 %. With the Checkpoint-2 (all-Petela) model, this station gives 22.8–23.0 %.

TABLE IV
HEADLINE RESULTS, NASA POWER 2021–2023 MEAN (MONTE CARLO 5–95 % RANGE IN BRACKETS)

Metric	Jodhpur	Bengaluru	Guwahati
POA irradiation [kWh/m^2/yr]	2,076	1,745	1,662
Diffuse + ground share [%]	37.1	51.6	48.4
Exergy content of sunlight Ex_{in}/E_{in} [–]	0.839	0.808	0.814
Energy-weighted T_c / max T_c [°C]	42.1 / 67.1	36.1 / 57.8	37.9 / 62.5
Specific yield [kWh/kWp] / PR [–]	1,883 / 0.863	1,591 / 0.867	1,508 / 0.862
Temperature loss / low-light loss [%]	5.0 / 1.5	3.2 / 2.2	3.8 / 2.4
η_{PV} [%]	20.83	20.94	20.82
ψ_{PV} [%]	24.83 (24.1–25.5)	25.92 (25.2–26.7)	25.57 (24.9–26.3)
ψ_{solar} / ψ_{sys} [%]	23.1 / 26.4	24.2 / 29.0	23.8 / 29.6
Solar coverage SCR [%]	79.7 (78.7–81.2)	74.3 (71.5–78.0)	70.4 (67.4–73.2)
of which direct PV / via battery [%]	49.8 / 29.9	48.5 / 25.7	45.0 / 25.4
Grid dependency [%] / export [MWh/yr]	20.3 / 22.5	25.7 / 13.4	29.6 / 12.7
PV share of exergy destroyed [%]	97.8	97.7	97.7
Battery EFC/yr / SOH after 1 yr	268 / 0.987	231 / 0.991	227 / 0.991
LCOE PV only / PV + battery [INR/kWh]	3.48 / 7.14	4.12 / 8.45	4.34 / 8.92
Energy tariff [INR/kWh]	6.00	4.50	8.72
NPV PV only / PV + battery [lakh INR]	+8.0 / −17.0	−2.5 / −31.8	+11.6 / −8.8
CO_2 avoided [t/yr]	61.8	52.2	49.5

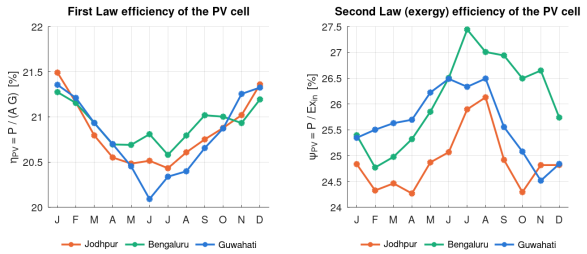


Fig. 3. Monthly η_{PV} (left) and ψ_{PV} (right). In the monsoon the energy efficiency falls while the exergy efficiency rises.

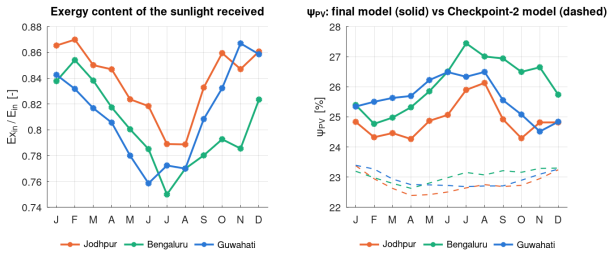


Fig. 4. Left: exergy content of the light received. Right: ψ_{PV}, final model (solid) vs Checkpoint-2 all-Petela model (dashed).

TABLE V
WINTER VS MONSOON (NASA POWER 2021–2023)

Season	Diffuse [%]	Ex_{in}/E_{in}	η_{PV} [%]	ψ_{PV} [%]
Jodhpur, winter	27	0.866	21.33	24.64
Jodhpur, monsoon	49	0.809	20.58	25.45
Bengaluru, winter	38	0.840	21.21	25.26
Bengaluru, monsoon	67	0.773	20.81	26.94
Guwahati, winter	36	0.845	21.30	25.21
Guwahati, monsoon	63	0.779	20.40	26.17

VI. RESULTS

A. PV performance, and what diffuse light does to the Second Law

Table V and Fig. 3 show four effects:

1) **The light itself is worth less in the monsoon.** Ex_{in}/E_{in} falls from ≈0.85 in winter to 0.77–0.81, as the diffuse share rises from about a third to about two thirds (Fig. 4).
2) η_{PV} **falls by 0.4–0.9 points**, from weak-light losses and warm air.
3) ψ_{PV} **rises by 0.8–1.7 points.** The input exergy per kWh of light falls 7–8 %, while the output per kWh falls only 2–4 %. By (22), ψ must rise. The cell wastes a smaller share of a smaller, lower-grade input; this is not a sign of better performance.
4) **Across cities,** ψ_{PV} **follows diffuse share, not sunshine.** Bengaluru (52 % diffuse) is highest at 25.92 %, then Guwahati (48 %) at 25.57 %, then Jodhpur (37 %, hottest cells) at 24.83 %. Check: $20.83/0.839 = 24.83$.

Hour by hour, ψ_{PV} climbs from ≈24 % under clear skies to ≈29 % under full overcast (Fig. 5).

What each model change did (Table VI). The diffuse-exergy split raises ψ_{PV} by 2.5–3.5 points and opens a 1.1-point spread between cities, against 0.3 points in the old model. The low-light factor lowers yield by 1.5–2.4 % and ψ_{PV} by 0.3–0.6 points. The Checkpoint-2 conclusion that climate leaves conversion quality unchanged came from the old model's assumptions.

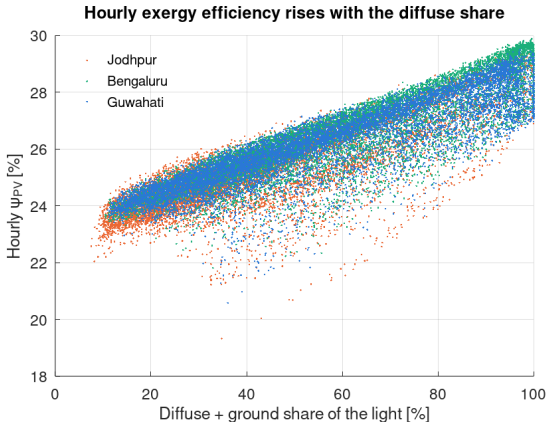


Fig. 5. Hourly ψ_{PV} against the diffuse share. Points below the band are low-sun hours (weak light, steep incidence).

TABLE VI
ANNUAL ψ_{PV} [%] UNDER EACH PV-MODEL VARIANT (CP2 = CHECKPOINT 2: PETELA FACTOR FOR ALL LIGHT, NO LOW-LIGHT TERM)

Variant	Jodhpur	Bengaluru	Guwahati
Checkpoint 2 model	22.75	23.03	22.94
+ split exergy only	25.22	26.52	26.21
+ low-light only	22.40	22.50	22.38
Final (both)	**24.83**	**25.92**	**25.57**
Yield, CP2 [kWh/kWp]	1,913	1,629	1,546
Yield, final [kWh/kWp]	1,883	1,591	1,508

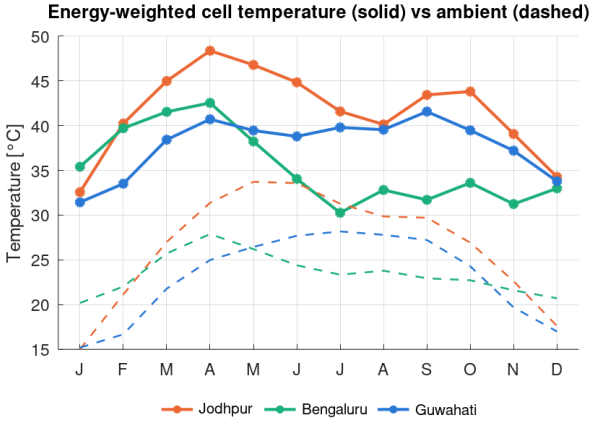


Fig. 6. Monthly energy-weighted cell temperature vs ambient.

Cell temperature (Fig. 6). Jodhpur's cells run hottest on an energy-weighted basis: 42.1 °C, against 36.1 °C in Bengaluru and 37.9 °C in Guwahati. This costs 5.0 %, 3.2 % and 3.8 % of output. While producing, cells run 7–19 K above ambient on a monthly basis. The NOCT correlation would run them 6.0–9.0 K hotter and under-predict yield by 1.8–2.8 %.

B. Where the exergy is destroyed

In every city, 23–24 % of the incident solar exergy reaches the EV, the grid or the battery. The other 76–77 % is destroyed

TABLE VII
EXERGY DESTROYED OR LOST [KWH/YR, 3-YR MEAN]

	Jodhpur	Bengaluru	Guwahati
Solar exergy in Ex_{in}	376,253	304,568	292,591
PV internal	226,411	179,819	172,515
PV optical	47,103	39,589	37,620
PV rejected heat	9,325	6,224	7,650
Inverter	2,746	2,321	2,199
DC wiring	1,868	1,579	1,496
Battery PCS	1,527	1,314	1,296
Battery I^2R	209	164	162
Total	**289,189**	**231,010**	**222,939**

TABLE VIII
SOLAR COVERAGE BY SEASON [%]

	Jodhpur	Bengaluru	Guwahati
Winter (DJF)	76.6	76.4	70.5
Pre-monsoon (MAM)	84.7	80.6	72.1
Monsoon (JJAS)	79.3	71.6	67.7
Post-monsoon (ON)	77.6	67.0	72.8
CoV of monthly SCR	0.055	0.095	0.063

or lost, and 98 % of that happens in the PV cell (Fig. 7, Table VII).

- **The hypothesis holds, but it was close to guaranteed by structure.** The power-electronic components have $\psi = \eta \approx 0.97$–0.98. Only the PV cell turns a non-work input into work, and it converts about a quarter of it. The informative result is inside the PV term: about 80 % is internal destruction, 17 % optical loss and only 3 % the work potential of the rejected heat. That heat leaves only 7–19 K above T_0, so its Carnot factor is a few per cent. Cooling a cell pays through β, not by recovering heat.
- **With all-Petela input**, PV destruction would be 14–20 % higher, because cloud light would be credited with work potential it never had.
- **The small terms rank the same in every city:** inverter > wiring > PCS > battery I^2R. The inverter destroys 13–14 times more than the cells' resistance.
- **The battery is a near-reversible store.** Its I^2R destruction is 162–209 kWh/yr, ≈0.07 % of the total. The pack never runs more than 6.3–6.9 K above ambient, and S_{gen} is 1,960–2,489 kJ/K per year. Fade is 0.9–1.3 %/yr, highest in Jodhpur (268 cycles/yr, hottest). Tripling R to cover DC versus AC resistance raises battery destruction to 488–631 kWh/yr and lowers coverage by 0.2 points.

C. Coverage, seasons and grid dependency

Annual coverage is 79.7 %, 74.3 % and 70.4 %. PV supplies 45–50 % of demand directly and the battery a further 25–30 %.

Weak seasons. Each city has a different weak season (Fig. 8, Table VIII):

- Guwahati: the monsoon (58.6 % in June).
- Bengaluru: the north-east monsoon (62.2 % in November).

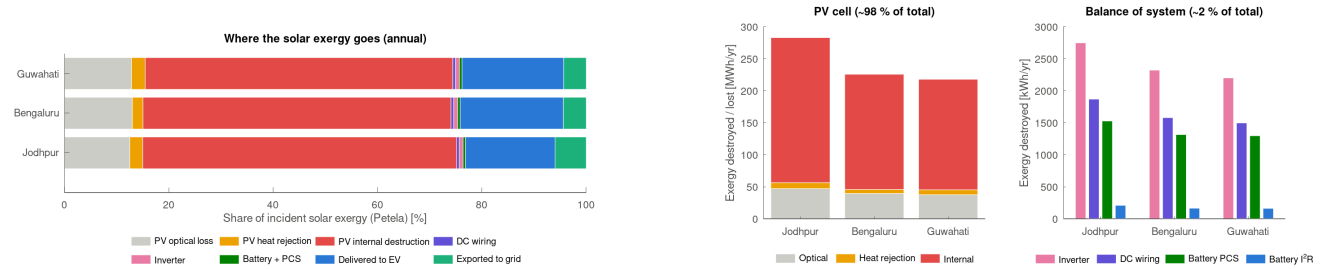


Fig. 7. Left: fate of the solar exergy. Right: exergy destruction by subsystem; note the MWh vs kWh scales. (The left panel's axis label reads "Petela" from the plotting script, but the bars are computed with the final split-exergy model.)

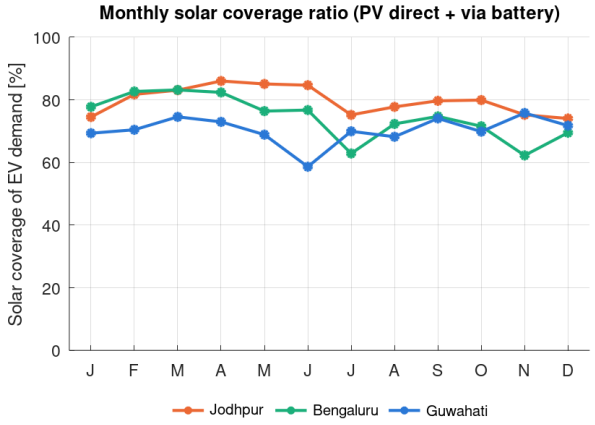


Fig. 8. Monthly solar coverage.

- Jodhpur: winter (74.0 % in December), when the evening peak falls in darkness.

Jodhpur, not Bengaluru, has the flattest year. Even the best month reaches only 86 %: with measured timing, the limit is the evening and night load, not sunshine.

Daily dispatch (Fig. 9). PV covers ≈08:00–16:00 and fills the battery. The battery carries the 17:00–22:00 peak, and the grid supplies the night and the morning ramp. In Jodhpur the battery usually lasts to midnight; in Guwahati it runs out around 21:00. Between 75 % and 82 % of PV output is used on site.

D. Sensitivity analysis

Each input in Table IX is changed one at a time, with everything else at the baseline (Fig. 10). The coverage ranking survives every change except the weather source. The exergy findings survive all of them.

E. Uncertainty: weather years and Monte Carlo

Year to year. Coverage in 2021/2022/2023 was 79.6/80.4/80.2 % in Jodhpur, 73.5/73.3/76.8 % in Bengaluru and 70.9/68.7/72.1 % in Guwahati. The year 2023 was the best for Bengaluru and Guwahati, so a single-year study would read them ≈2 points high. Every ranking holds in every year (Fig. 12).

Monte Carlo. Each of 200 draws picks one weather year and samples six inputs uniformly (Table X). The same draw is applied to all three cities (paired), so the city-to-city comparison always shares identical assumptions. The design, the tariffs and the emission factor are not sampled. Results are in Table XI and Fig. 11.

The city ranges overlap for ψ_{PV} only because each draw shifts all three cities together; within every draw, the order is the same. The median NPV sits below the baseline because the battery price range (16–25k) is centred above the 18k baseline. The economics are the least certain output: the battery price alone moves NPV by about ±6 lakh.

F. Economics and CO_2

- **Battery cost.** Capex is 49.8 lakh: PV 19.8, battery 16.8, converters 6.7 and installation 6.5. The PV-only station costs 26.2 lakh. The battery is 34 % of capex and buys 25–30 points of coverage. It also roughly doubles the LCOE, because each kWh shifted pays for storage and loses ≈7 % in two PCS passes.
- **PV-only LCOE** is below the tariff in all three cities (Fig. 13). The PV-only NPV is nevertheless *negative* in Bengaluru (−2.5 lakh). About half of PV output is exported at 3 INR/kWh, and Karnataka's 4.50 INR/kWh tariff leaves little margin even on the self-consumed half.
- **Guwahati has the least sun and the best economics.** Assam's 8.72 INR/kWh tariff gives a PV-only payback of 7.0 years. With the battery, LCOE is 8.92 INR/kWh against an 8.72 tariff, and NPV is −8.8 lakh, negative in all 200 draws (best case −6.3 lakh).
- **CO_2.** Avoided CO_2 is 61.8/52.2/49.5 t/yr. It can exceed the grid-only total of 57.5 t/yr because exports also displace grid power. City differences come only from solar output, since a single national factor is used.
- **The economic ranking is set by tariff, not climate.** Fixed and demand charges are excluded because the evening EV load sets contract demand either way.

VII. Expected Patterns vs Findings

Table XII compares the Checkpoint-2 expectations with the results. The main correction is that monsoon cloud was expected to degrade Second-Law performance, and it does the opposite. Cloud reduces both the *quantity* of solar exergy and its *quality* (0.67–0.80 for diffuse light against 0.93 for the beam), and the panel converts low-quality light with a higher

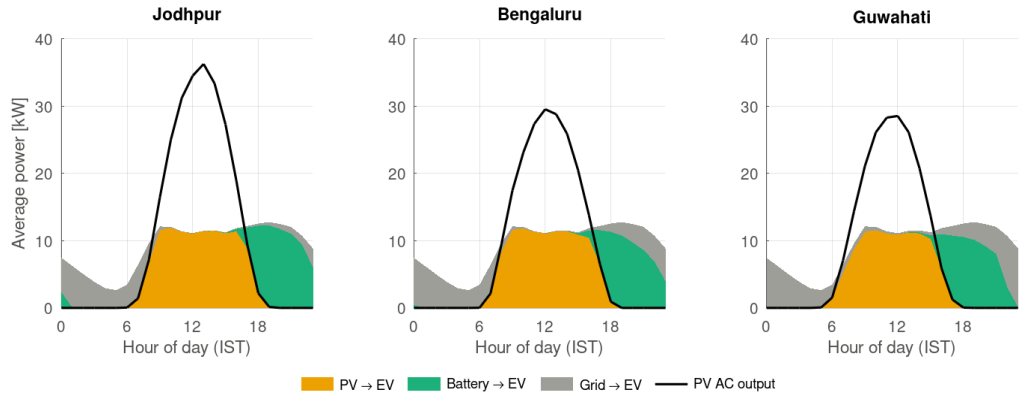


Fig. 9. Average daily dispatch, 3-year mean.

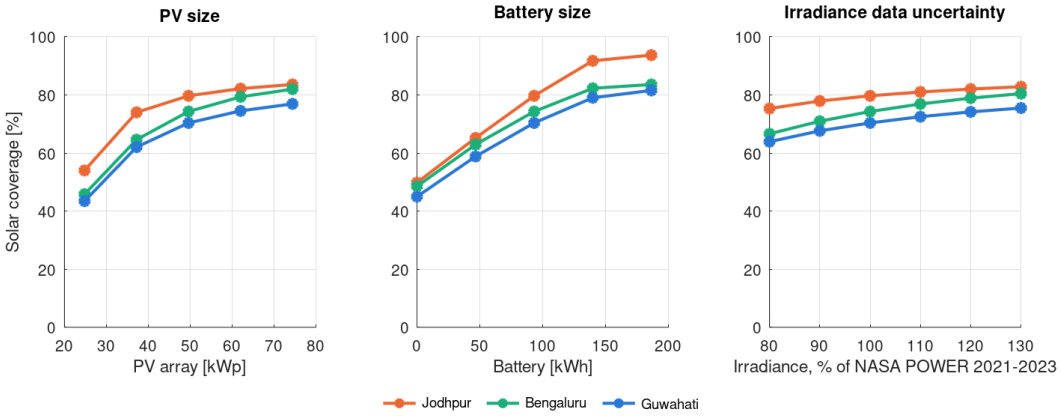


Fig. 10. Coverage vs PV size, battery size and irradiance scale.

TABLE IX
ONE-AT-A-TIME SENSITIVITY

Change	Effect
Weather: PVGIS-ERA5 2023 instead of NASA POWER	Coverage 71.8/76.6/61.7 % (J/B/G): Bengaluru moves to first. ψ_{PV} 24.56/24.89/24.94 %: smaller spread because ERA5 has less diffuse light, but monsoon ψ is still above winter ψ everywhere. The subsystem ranking is unchanged.
EV timing: workplace / residential / synthetic	Jodhpur 77.2/61.4/89.9 %, Bengaluru 71.9/59.1/81.4 %, Guwahati 66.7/55.0/78.6 %. Evening-heavy timing costs 15–18 points; the synthetic profile overstated coverage by 7–10 points. City order unchanged.
Irradiance 80→130 %	Coverage moves only 12–14 points over the whole range; Jodhpur stays first even at 80 %. ψ_{PV} moves by $\leq$0.2 points.
PV array 24.8→74.4 kWp	Coverage 54→84 % (J), 46→82 % (B), 43→77 % (G). Above 49.6 kWp most extra PV is exported (39–48 % of output at 74.4 kWp).
Battery 0→186 kWh	+25–30 points up to 93 kWh, +8–12 more at 140 kWh, then +1–3. With measured timing the knee is $\approx$140 kWh.
Cell temperature: NOCT correlation	Cells 6.0–9.0 K hotter, yield 1.8–2.8 % lower; ranking unchanged.
Battery resistance $\times$3	Battery exergy destruction $\times$3 (488–631 kWh/yr); coverage $-$0.2 points.
PV model: Checkpoint 2 vs final	ψ_{PV} changes by 2.1–2.9 points and its city ranking changes; yield and coverage change by $\approx$2 %.

exergy efficiency. An energy-only analysis cannot see this; it shows only that cloudy hours produce less.

VIII. ASSUMPTIONS AND THEIR DEFENCE

Table XIII (Appendix B) lists every assumption that a result depends on. For each one it gives the source and the defence,

TABLE X
MONTE CARLO INPUTS (UNIFORM, PAIRED ACROSS CITIES)

Input	Range	Basis
Irradiance scale	0.93–1.07	POWER GHI ≈0 bias vs BSRN; ±7 % for 1° cell + transposition
Module η_{ref}	±3 %	Datasheet P_{max} tolerance
NOCT	41–45 °C	Datasheet 43±2 (h_{free} recalibrated)
ν_{inv}, ν_{pcs}	0.96–0.98 each	Project brief
Cell R	0.30–0.90 mΩ	1 kHz AC value; DC is 2–3×
Battery price	16–25k INR/kWh	2026 C&I LFP quotes [30]
Weather year	2021/22/23	Equal probability

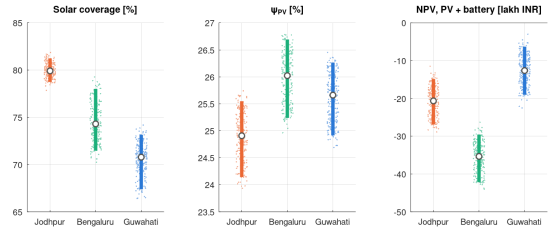


Fig. 11. Monte Carlo: 5–95 % bars, median (circle) and individual draws.

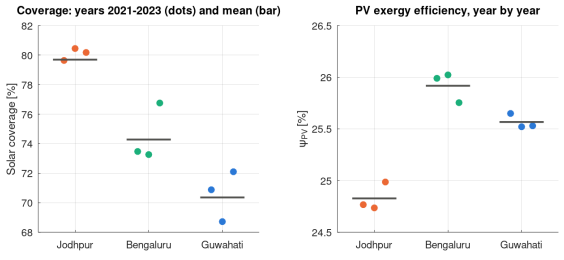


Fig. 12. Coverage and ψ_{PV} in each weather year.

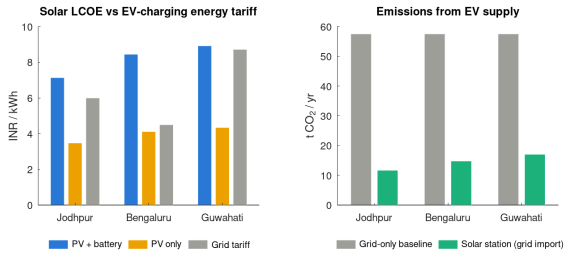


Fig. 13. LCOE vs tariff, and CO_2 vs the grid-only baseline.

and it says where the defence is weak.

IX. LIMITATIONS

- **Weather source.** NASA POWER and ERA5 differ by 21 % (Jodhpur) and 24 % (Guwahati) in 2023, and the coverage ranking flips between them. This is the largest single uncertainty, and it lies outside the Monte Carlo. A higher-resolution satellite set (NSRDB, Meteosat-IODC) is the next check.
- **EV timing** comes from Korean chargers and repeats identically every day.
- **Diluted-radiation exergy** rests on a grey, isotropic simplification. The environment term for diluted light is debated in the literature.
- **Not modelled:** soiling (2–5 %/yr in Rajasthan dust), inverter part-load efficiency, spectral effects and calendar ageing. The performance ratio of 0.86–0.87 is therefore optimistic.
- **Three years** of weather are not a climatology, and the Monte Carlo ranges are uniform and independent. Its results are bounds on plausible outcomes, not calibrated probabilities.
- **Open check:** the PVGIS PVcalc yield comparison (49.6 kWp, 20°/180°) for each site.

X. CONCLUSION

One fixed 49.6 kWp/93.2 kWh station covers 70–80 % of a measured-timing EV load in three Indian climates. Coverage is set more by evening demand and battery size than by the last 10 % of sunshine. The Second Law shows what the First Law hides:

1) The PV cell destroys ≈98 % of the exergy, mostly internally.
2) The rest of the chain is nearly reversible: inverter > wiring > PCS ≫ cells.
3) Diffuse light lowers both the quantity and the quality of the solar input, so ψ_{PV} rises in cloudy weather and the sunniest city has the lowest exergy efficiency.

The economic verdict follows the tariff, not the thermodynamics: PV alone generates below the tariff everywhere, and the battery has a negative NPV everywhere at current prices. Station sizing and the storage decision should therefore be made per city, chiefly on tariff and expected EV timing.

REPRODUCIBILITY

All code, data and outputs are in the project folder (`thermo_project`). The main case is `main` (NASA POWER 2021–2023, sensitivity and Monte Carlo, ≈25 min). The cross-check is `source='ERA5'; main` (≈3 min). The energy and exergy closures are asserted on every run.

ACKNOWLEDGEMENT

[Add acknowledgements for the course instructor and TA here.]

REFERENCES

[1] R. Petela, "Exergy of heat radiation," *J. Heat Transfer*, vol. 86, no. 2, pp. 187–192, 1964.
[2] P. T. Landsberg and G. Tonge, "Thermodynamics of the conversion of diluted radiation," *J. Phys. A: Math. Gen.*, vol. 12, no. 4, pp. 551–562, 1979.
[3] T. Huld, G. Friesen, A. Skoczek, R. P. Kenny, T. Sample, M. Field and E. D. Dunlop, "A power-rating model for crystalline silicon PV modules," *Sol. Energy Mater. Sol. Cells*, vol. 95, no. 12, pp. 3359–3369, 2011.
[4] A. Chatzipanagi *et al.*, "An updated simplified energy yield model for recent photovoltaic module technologies," *Prog. Photovolt. Res. Appl.*, vol. 33, no. 8, pp. 905–917, 2025.

TABLE XI
MONTE CARLO RESULTS (5 % / MEDIAN / 95 %)

	Jodhpur	Bengaluru	Guwahati
SCR [%]	78.7/79.9/81.2	71.5/74.3/78.0	67.4/70.8/73.2
ψ_{PV} [%]	24.1/24.9/25.5	25.2/26.0/26.7	24.9/25.7/26.3
LCOE [INR/kWh]	6.8/7.5/8.4	7.8/8.9/10.2	8.3/9.4/10.6
NPV [lakh]	−26.9/−20.6/−14.8	−42.2/−35.3/−29.6	−19.0/−12.6/−6.3

Share of draws: coverage J>B>G 100 %; ψ_{PV} monsoon>winter 100 % (each city); ψ_{PV} of Jodhpur lowest 100 %; NPV>0 0 %; Guwahati best NPV 100 %.

TABLE XII
CHECKPOINT-2 EXPECTATIONS VS RESULTS

Expected	Result (NASA POWER 2021–2023)	Verdict
Coverage J > B > G	80 > 74 > 70 %, every year, 100 % of draws; ERA5 puts Bengaluru first	Supported; weather-source sensitive
ψ: Jodhpur highest with summer dip; Bengaluru flattest; Guwahati lowest with monsoon drop	ψ_{PV} B 25.9 > G 25.6 > J 24.8 %; ψ_{PV} *rises* in the monsoon everywhere	Reversed
PV cell dominates exergy destruction	97.7–97.8 % in all cities	Supported (structural)
Guwahati monsoon dip; Jodhpur heat hit; Bengaluru flattest	Guwahati weakest in monsoon; Jodhpur hottest cells (5.0 % loss); Jodhpur flattest (CoV 0.055 vs 0.095)	Partly
Grid dependency, cost, CO_2 opposite to coverage	Grid dependency and CO_2 yes; cost no (Guwahati best NPV via tariff)	Partly
No single national verdict	Rankings differ by metric; tariff decides economics; EV timing moves coverage as much as climate	Supported

[5] J. A. Duffie and W. A. Beckman, *Solar Engineering of Thermal Processes*, 4th ed. Hoboken, NJ: Wiley, 2013.

[6] J. E. Hay and J. A. Davies, "Calculation of the solar radiation incident on an inclined surface," in *Proc. First Canadian Solar Radiation Data Workshop*, 1980, pp. 59–72.

[7] N. Martin and J. M. Ruiz, "Calculation of the PV modules angular losses under field conditions by means of an analytical model," *Sol. Energy Mater. Sol. Cells*, vol. 70, pp. 25–38, 2001.

[8] A. F. Souka and H. H. Safwat, "Determination of the optimum orientations for the double-exposure, flat-plate collector and its reflectors," *Solar Energy*, vol. 10, no. 4, pp. 170–174, 1966.

[9] W. C. Swinbank, "Long-wave radiation from clear skies," *Q. J. R. Meteorol. Soc.*, vol. 89, pp. 339–348, 1963.

[10] J. H. Watmuff, W. W. S. Charters and D. Proctor, "Solar and wind induced external coefficients for solar collectors," *COMPLES*, no. 2, p. 56, 1977.

[11] A. D. Jones and C. P. Underwood, "A thermal model for photovoltaic systems," *Solar Energy*, vol. 70, no. 4, pp. 349–359, 2001.

[12] M. K. Fuentes, "A simplified thermal model for flat-plate photovoltaic arrays," Sandia National Laboratories, SAND85-0330, 1987.

[13] D. L. Evans, "Simplified method for predicting photovoltaic array output," *Solar Energy*, vol. 27, no. 6, pp. 555–560, 1981.

[14] "Thermal and electrical model of a PV module with radiative and convective heat transfer," *Thermal Science*, 2019, paper TSCI180512020P (glass 0.85, backsheet 0.91, Swinbank sky). [Online]. Available: https://thermalscience.vinca.rs/pdfs/papers-2019/TSCI180512020P.pdf

[15] A. Bejan, *Advanced Engineering Thermodynamics*, 4th ed. Hoboken, NJ: Wiley, 2016.

[16] J. Wang *et al.*, "Cycle-life model for graphite-$LiFePO_4$ cells," *J. Power Sources*, vol. 196, no. 8, pp. 3942–3948, 2011.

[17] F. Bayrak, N. Abu-Hamdeh, K. A. Alnefaie and H. F. Öztop, "A review on exergy analysis of solar electricity production," *Renew. Sustain. Energy Rev.*, vol. 74, pp. 755–770, 2017.

[18] "Explainable AI-enabled adaptive fuzzy MPPT and energy management for bifacial PV and battery-powered EV charging system," 2026. [Online]. Available: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12873354/

[19] "Techno-economic analysis and optimized PV-powered EV charging facilities under various climate conditions in India," *Discover Sustainability*, 2025, doi:10.1007/s43621-025-01119-7.

[20] "Techno-economic and policy analysis of solar-wind powered EV charging for sustainable e-mobility," *Sci. Rep.*, 2026, doi:10.1038/s41598-026-49453-8.

[21] NASA Langley Research Center, POWER Project, hourly API (CERES SYN1deg, MERRA-2), 2021–2023. [Online]. Available: https://power.larc.nasa.gov

[22] European Commission JRC, PVGIS hourly time series, PVGIS-ERA5, 2023. [Online]. Available: https://re.jrc.ec.europa.eu/pvg_tools/

[23] "A dataset for multi-faceted analysis of electric vehicle charging transactions," *Sci. Data*, vol. 11, 2024, doi:10.1038/s41597-024-02942-9; data doi:10.6084/m9.figshare.22495141 (CC BY 4.0).

[24] Trina Solar, "Vertex N TSM-NE19R datasheet," DT-M-0093-B, 2024.

[25] REPT BATTERO, "Energy storage battery solution" brochure, 2026.

[26] PVsyst SA, "Array ohmic wiring loss," PVsyst 7 Help. [Online]. Available: https://www.pvsyst.com/help/ohmic_loss.htm

[27] A. P. Dobos, "PVWatts version 5 manual," NREL/TP-6A20-62641, 2014.

[28] Central Electricity Authority, "CO_2 Baseline Database for the Indian Power Sector, User Guide v21.0," 2025.

[29] APDCL tariff notice FY2026–27 (LT-X EV charging, 8.72 INR/kWh); KERC 2025 order LT-6(c) (4.50 INR/kWh); RERC EV/LT public charging (6.00 INR/kWh).

[30] "C&I BESS prices in India, 2026." [Online]. Available: https://lithiuminverter.in/prices/industrial-bess.html

APPENDIX A
DERIVATIONS

A. Petela's exergy of blackbody radiation

For isotropic blackbody radiation at temperature T, the hemispherical energy flux is $E = \sigma T^4$ and the entropy flux is $S = \frac{4}{3}\sigma T^3$. (In the radiation field, $s = \frac{4}{3}u/T$ per unit volume and $u = aT^4$, with $\sigma = ac/4$.) The environment at T_0 radiates $E_0 = \sigma T_0^4$ and $S_0 = \frac{4}{3}\sigma T_0^3$. The exergy flux relative to that dead state is

$$Ex = (E - E_0) - T_0(S - S_0) = \sigma T^4 - \sigma T_0^4 - \tfrac{4}{3}\sigma T_0 T^3 + \tfrac{4}{3}\sigma T_0^4 = \sigma T^4 - \tfrac{4}{3}\sigma T_0 T^3 + \tfrac{1}{3}\sigma T_0^4.$$

Dividing by $E = \sigma T^4$ gives $\psi_P = 1 - \frac{4}{3}(T_0/T) + \frac{1}{3}(T_0/T)^4$. With $T = 5778\,\mathrm{K}$ and $T_0 = 300\,\mathrm{K}$, $T_0/T = 0.05192$, so $\psi_P = 1 - 0.06923 + 0.0000024 = 0.9308$.

B. Diluted radiation

Radiation with spectral radiance $\varepsilon B_\nu(T_s)$ has photon occupation $n = \varepsilon/(e^x - 1)$, $x = h\nu/kT_s$. The entropy of a boson mode is $k[(1+n)\ln(1+n) - n\ln n]$. Integrating over frequency and normalising by the undiluted case gives $S = \frac{4}{3}\varepsilon\sigma T_s^3 X(\varepsilon)$ with X as in the main text, $X(1) = 1$, and X increasing as ε falls. With $E = \varepsilon\sigma T_s^4$:

$$\psi = 1 - \frac{T_0 S}{E} = 1 - \tfrac{4}{3}\frac{T_0}{T_s}X(\varepsilon).$$

The Petela environment term is left out for diluted light. It is below 10^{-5} for the beam, and it is not well defined when the diluted radiation's effective temperature approaches T_0.

C. Why $\psi_{PV} > \eta_{PV}$ and why it rises with diffuse light

From $\eta = P/(AG)$ and $\psi = P/Ex_{in}$, $\psi/\eta = AG/Ex_{in} = 1/\bar{\psi}_{light}$, where $\bar{\psi}_{light}$ is the irradiance-weighted exergy factor of the received light: ≈ 0.93 for pure beam and ≈ 0.70 for diffuse. More diffuse light lowers $\bar{\psi}_{light}$ faster than weak-light and temperature effects lower η, so ψ rises. Jodhpur annual: $20.83/0.839 = 24.83\,\%$; Bengaluru $20.94/0.808 = 25.92\,\%$.

D. NOCT correlation as a linearisation

Linearise (6) with $P = 0$ and lump convection and radiation into one coefficient U_L: $\alpha G = U_L(T_c - T_a)$, so $T_c - T_a = (\alpha/U_L)G$. Setting $T_c = \mathrm{NOCT}$ at $800\,\mathrm{W/m^2}$ and $20\,^\circ\mathrm{C}$ fixes $\alpha/U_L = (\mathrm{NOCT} - 20)/800$, giving $T_c = T_a + (\mathrm{NOCT} - 20)G/800$. This form has three gaps: it ignores the electrical output (a producing panel at NOCT conditions runs $\approx$37 °C, not 43 °C), wind other than 1 m/s, and the T^4 sky exchange. That is why it runs 6–9 K hot.

E. Battery: Gouy–Stodola and power limits

Take a control volume around the pack and its immediate surroundings, in steady state over a step. The electrical exergy I^2R is degraded to heat $\dot{Q}$ at T_b, which crosses to T_0.

- Generation inside the pack: $\dot{Q}/T_b$.
- Generation across the $T_b \to T_0$ gradient: $\dot{Q}(1/T_0 - 1/T_b)$.
- Total: $\dot{S}_{gen} = \dot{Q}/T_0$, so $\dot{E}x_{dest} = T_0\dot{S}_{gen} = I^2R$.

The heat at T_b carried only $\dot{Q}(1 - T_0/T_b) \approx 2\,\%$ of $\dot{Q}$ as exergy, which is lost anyway.

With $I = 1000P/V$, the loss is kP^2 with $k = 1000R/V^2$. For charging, the stored gain $(P - kP^2)\Delta t$ must not exceed the headroom h, so $P_{max} = [1 - \sqrt{1 - 4kh/\Delta t}]/(2k)$. For discharging, $(P + kP^2)\Delta t \le a$, so $P_{max} = [-1 + \sqrt{1 + 4ka/\Delta t}]/(2k)$. Both are capped at 0.5C.

Series vs parallel. For N_s cells in series and N_p strings, $V = N_s V_c$, $R = N_s R_c/N_p$ and $I = P/V$. Then $I^2R = P^2 R_c/(N_s N_p V_c^2)$, which depends only on the cell count $N_s N_p$. At fixed total cell count, the layout does not change cell loss.

F. Hay–Davies transposition

$$G_b = \mathrm{DNI}\max(\cos\theta, 0), \qquad G_d = \mathrm{DHI}\left[A_i R_b + (1 - A_i)\tfrac{1+\cos\beta}{2}\right], \qquad G_r = \rho\,\mathrm{GHI}\,\tfrac{1-\cos\beta}{2},$$
$$A_i = \mathrm{DNI}/I_0, \quad R_b = \cos\theta/\cos\theta_z, \quad I_0 = 1367[1 + 0.033\cos(360n/365)], \quad \rho = 0.2.$$

Below 5° solar elevation, only the isotropic diffuse term is kept.

G. Worked hour (Jodhpur, clear noon)

Take $G_{\mathrm{eff}} = 900\,\mathrm{W/m^2}$, $T_a = 35\,^\circ\mathrm{C}$ and $v_{10} = 3\,\mathrm{m/s}$, so $v = 2.53\,\mathrm{m/s}$ and $h = 13.1\,\mathrm{W/m^2K}$. The coupled balance gives $T_c = 49.7\,^\circ\mathrm{C}$ (56.7 °C at $v_{10} = 0.5\,\mathrm{m/s}$). Then $f_{low} \approx 1.00$ and $\eta = 0.2295 \times [1 - 0.0029 \times 24.7] = 0.2131$. This gives $P_{DC} = 0.2131 \times 216.1 \times 900 = 41.4\,\mathrm{kW}$ and $P_{AC} = 41.4 \times 0.98 \times 0.97 = 39.4\,\mathrm{kW}$, with 0.8 kW lost in wiring and 1.2 kW in the inverter.

APPENDIX B
ASSUMPTION REGISTER

TABLE XIII: Assumption register: value, origin, and defence

Assumption	Value	Origin	Defence (and known weakness)
Design and boundary			
Fixed design in all cities	–	Study design	Makes climate the only physical variable. Weakness: not cost-optimal per city; PV and battery sweeps are reported.
Module choice	Trina NE19R 620 W	Datasheet	Representative tier-1 N-type TOPCon with complete thermal and electrical data. An ALMM-listed Indian TOPCon module has near-identical β and NOCT, so the thermodynamic results transfer; only cost would change.
Array size	49.6 kWp	Energy sizing	Annual yield $\approx$ annual demand at 1,630 kWh/kWp. Not an economic optimum: PV-only LCOE is below the tariff, so a larger array may raise NPV.
Tilt / azimuth	20° / 180°	PVGIS download setting	Due south is optimal in the northern hemisphere. Re-transposition shows 20° loses 0.67/0.40/0.86 % of POA against the 28/14/28° optima. One tilt keeps geometry from being a second variable.
Battery chemistry	LFP	Literature, safety	No oxygen-releasing cathode, so safer thermal runaway behaviour; 8,000 cycles; no Co or Ni; same chemistry as the Gorakhpur station [18].
Battery size / layout	93.2 kWh, 1P104S	Two standard Y52 modules	Series string at 333 V matches the PCS window and halves cable current. Cell I^2R is the same as for 2P52S. 0.5C (46.6 kW) matches the PV peak. Knee is at $\approx$140 kWh; NPV is negative at every size.
SOC window	10–90 %	Engineering	Common practice for LFP longevity; 74.6 kWh usable.
EMS	Priority rule, PV-only charging	Scope	Transparent and reproducible; optimisation is out of scope. Grid charging would raise grid CO_2 and was excluded on purpose.
EV demand level	222 kWh/day at AC bus	Design value	Held fixed for like-for-like comparison. Charger conversion loss ($\approx$5 %) is outside the boundary: if 222 kWh is energy into car batteries, the bus demand would be $\approx$233 kWh.
EV timing	Korean public sessions	Measured, open data	No open Indian session data. Weakness: Indian highway and fleet charging may be more daytime-heavy (coverage up towards 79–90 %), residential more evening-heavy (down 15–18 points). Identical every day: no weekday or seasonal pattern.
Weather			
Main source	NASA POWER 2021–23	Satellite	Matches long-term solar maps and clear-sky physics; $\approx$0 GHI bias against BSRN. Weakness: 1° ($\approx$100 km) grid cell.
Transposition	Hay–Davies, albedo 0.2	Literature [6]	Standard anisotropic model; 0.2 is the PVsyst/PVGIS default. Covered by the $\pm$7 % Monte Carlo irradiance range.
ERA5 as cross-check only	–	Diagnosis	ERA5 under-reads clear-sky winter irradiance in north India by $\sim$50 % at noon. Weakness: the coverage ranking flips on ERA5, so the dataset choice dominates all model uncertainty.
PV optics and thermal			
Beam IAM	ASHRAE, $b_0 = 0.05$	Literature [8]	PVsyst default for glass-faced modules.
Diffuse / ground IAM	0.95 / 0.65	Martin & Ruiz [7] / estimate	0.95 from the effective-IAM approach at 20°. 0.65 has no direct source, but ground light is $<$1 % of POA at 20°, so it is immaterial.
Absorptance α	0.90	Duffie & Beckman [5]	Their NOCT model uses $\tau\alpha \approx 0.9$; papers range 0.9–0.97. The NOCT calibration of h_{free} absorbs errors at the test point.
Emissivities ε_f, ε_b	0.84, 0.89	Literature range	Published values: glass 0.84–0.91, backsheet 0.85–0.91 [14]. Calibration again compensates at the NOCT point.
Sky temperature	$0.0552T_a^{1.5}$	Swinbank [9]	Standard clear-sky correlation. Weakness: overcast skies are warmer, so radiative cooling is slightly overstated in the monsoon.

Assumption	Value	Origin	Defence (and known weakness)
Wind coefficient	3.0 W/m^{2}K per m/s	Watmuff [10]	Valid for 1–7 m/s. Weakness: Watmuff's full form $(2.8 + 3.0v)$ already contains free convection and radiation; here only the slope is reused and the constant is replaced by the calibrated h_{free}, with radiation modelled explicitly. At the NOCT point nothing is double-counted.
h_{free}	5.55 W/m^{2}K	Calibrated	Reproduces datasheet NOCT exactly. Weakness: higher than pure natural convection (2–4), because it absorbs errors in α and ε; it is a lumped calibrated coefficient.
Wind height scaling	$(3/10)^{1/7}$	Power law	Standard neutral-stability exponent; 3 m canopy height.
Steady state per hour	$dT_c/dt = 0$	Time constant	Module time constant 5–10 min $\ll$ 1 h.
PV electrical			
Temperature coefficient	-0.29 %/K	Datasheet	Measured by the manufacturer.
Low-light k_1, k_2	-0.017237, -0.040465	PVGIS-5 c-Si [3]	Standard published fit. Weakness: fitted to older modules; the PVGIS-6 refit [4] has $k_1 = -0.0068$, $k_2 = -0.0164$ ($\approx$40 %), so weak-light loss is overstated for TOPCon and monsoon η and coverage are slightly low.
Hybrid efficiency model	Huld irradiance + datasheet β	Choice	Huld's temperature terms were fitted to ≈ -0.47 %/K modules. Described as a hybrid, not "the Huld model".
DC wiring + mismatch	2 %	Industry defaults	PVsyst's typical STC ohmic loss of 1.5 % gives $\approx$0.8 % annual [26]; PVWatts and Aurora mismatch defaults are 2 % [27]; sum 1.8–2.8 %. Weakness: constant 2 % instead of $\propto I^2$.
Inverter efficiency	0.97	Brief (96–98 %)	About 1 point below typical weighted efficiencies of modern string inverters. Weakness: constant; real efficiency drops at part load.
Exergy			
Dead state	$T_0 = T_a$ hourly	Convention	Standard for ambient-coupled systems.
T_{sun}	5778 K	Constant	Effective blackbody temperature of the photosphere.
Diffuse exergy	Grey-diluted, isotropic	Landsberg & Tonge [2]	Physically correct direction. Weakness: real sky light is bluer, and the Hay–Davies circumsolar part is near-beam, so diffuse exergy is somewhat underestimated and ψ_{PV} overestimated on hazy hours. The size of the effect is uncertain; its direction is robust. Petela's environment term is omitted for diluted light.
Electricity exergy	$Ex = E$	Definition	Electrical work is pure exergy.
Battery			
Cell resistance	0.30 mΩ (0.30–0.90 in MC)	Listing (AC 1 kHz)	DC resistance is typically 2–3$\times$ AC, so it is propagated; $\times$3 changes coverage by only 0.2 points.
Ageing	3.75×10^{-5}/EFC, Arrhenius	Brochure + Wang [16]	Cycle ageing only; calendar ageing not modelled.
Thermal mass / UA	572 kg$\times$1100, 40 W/K	Assumption	Affects only T_b (max rise $\approx$7 K) and fade marginally.
Converter efficiency	PCS 0.97 each way	Brief	Same class of power electronics as the PV inverter.
Economics (reporting only)			
Prices	PV 40k/kWp, batt 18k/kWh, inv 6k/kW, PCS 8k/kW, BOS 15 %	2025–26 market, assumptions	Indicative; battery price propagated (16–25k).
Financial	8 % real, 25 yr, O&M 1.5 %	Assumption	Typical C&I. Life equals the module power warranty.
Replacements	Battery yr 15 at 60 %; converters yr 12	Assumption	Calendar-limited pack life; falling prices.
Export credit	3 INR/kWh	APPC-level assumption	Conservative; drives the Bengaluru PV-only NPV sign.
Tariffs	6.00 / 4.50 / 8.72	State EV tariffs	RERC, KERC, APDCL orders [29]; energy charge only.
Emission factor	0.710 t/MWh	CEA v21 [28]	Single national grid; no state factors published.